

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 19

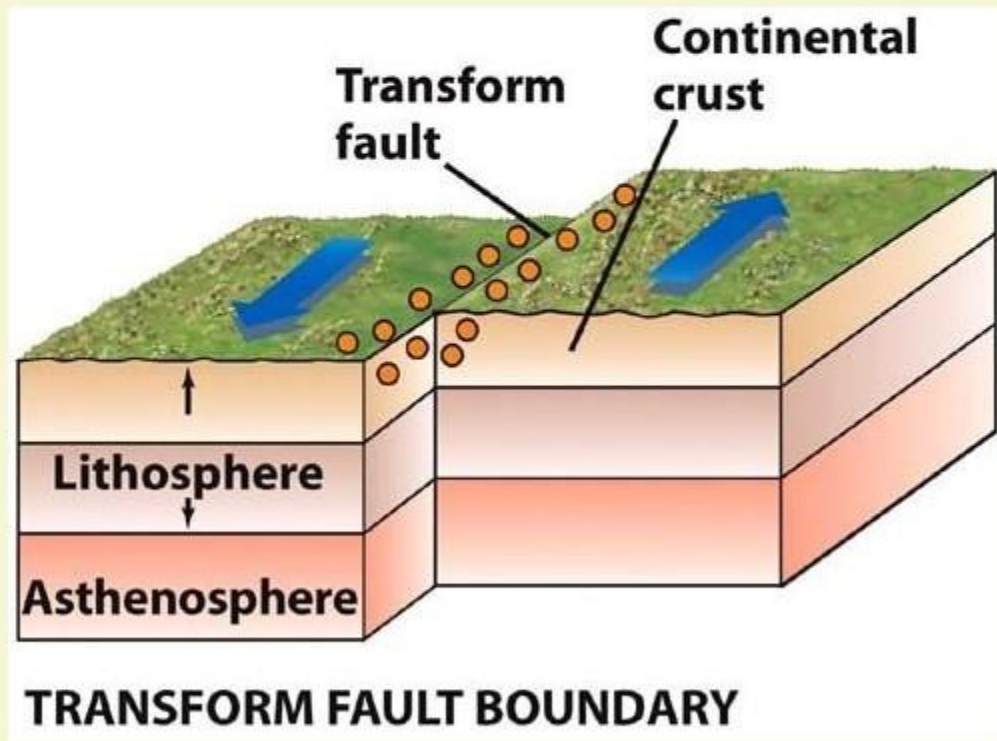
VOLCANOES

- **WORLD DISTRIBUTION OF VOLCANOES**

PLATE TECTONICS AND VULCANISM

MECHANISM AND CAUSES OF VULCANISM

WHAT HAPPENS AT TRANSFORM PLATE BOUNDARIES?



- 1 In the case of transform plate boundaries, the **plates slide past one another** without any collision or subduction or divergence.
- 2 Hence there is no gap created for release of pressure, gases or molten magma.
- 3 Thus, there is **no volcanic activity** associated with transform boundaries.
- 4 However as the plates rub against each other, **huge stresses are set up** that can cause portions of the rock to break, resulting in **earthquakes**. Example: San Andreas boundary in California.

DISTRIBUTION OF VOLCANOES IN THE WORLD

BELTS OF VOLCANOES IN THE WORLD



There are 3 major belts or zones of volcanoes in the world. They are:

- I. Circum-Pacific Belt
- II. Mid-Continental Belt
- III. Mid-Oceanic Ridge Belt



DISTRIBUTION OF VOLCANOES IN THE WORLD

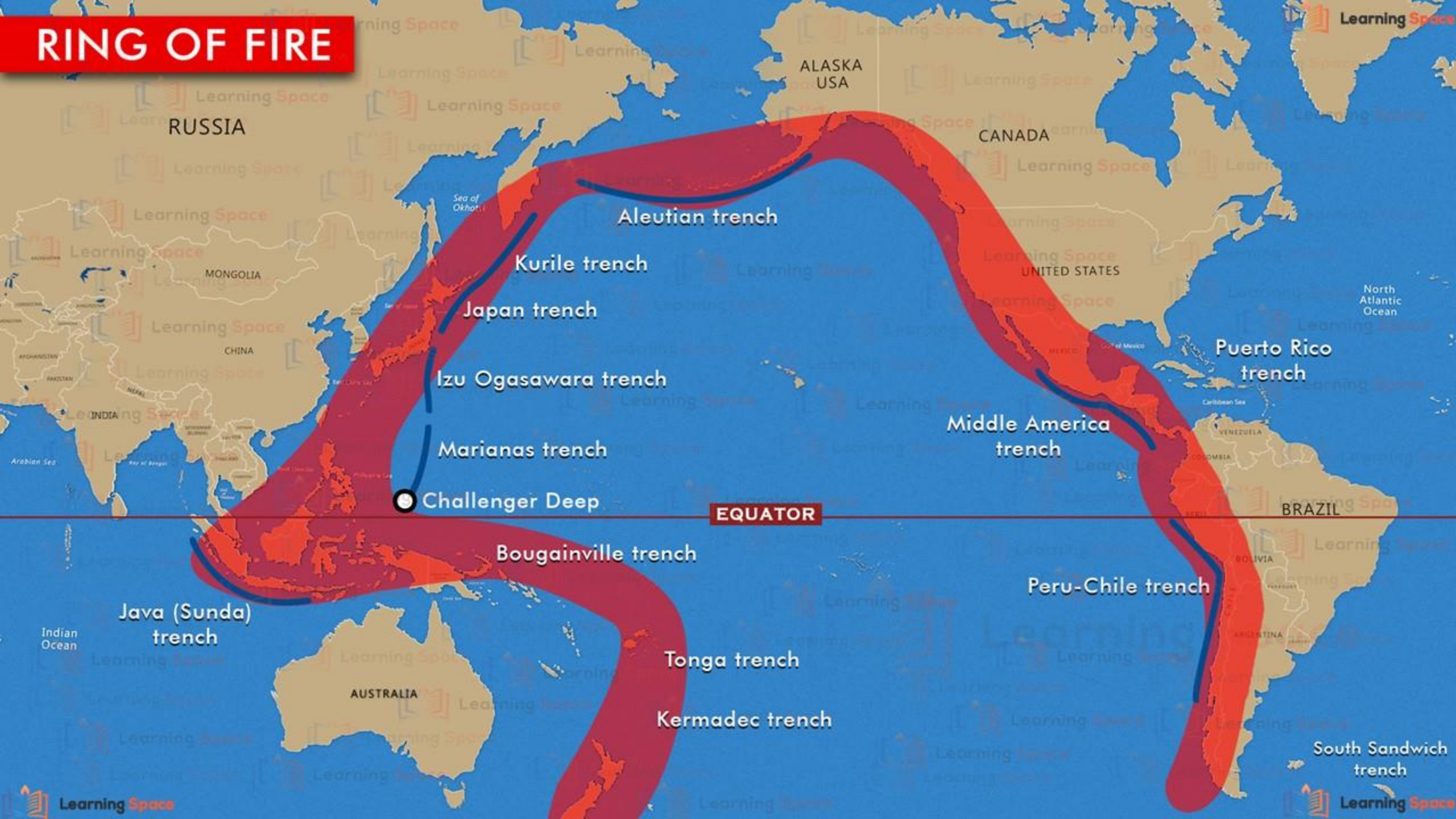
1 CIRCUM-PACIFIC BELT

1 The circum-Pacific Belt, also known as the 'volcanic zones of the convergent oceanic plate margins', includes the volcanoes of the eastern and western coastal areas of the Pacific Ocean (or the western coastal margins of North and South Americas and the eastern coastal margins of Asia), of island arcs and festoons off the east coast of Asia and of the volcanic islands scattered over the Pacific Ocean.

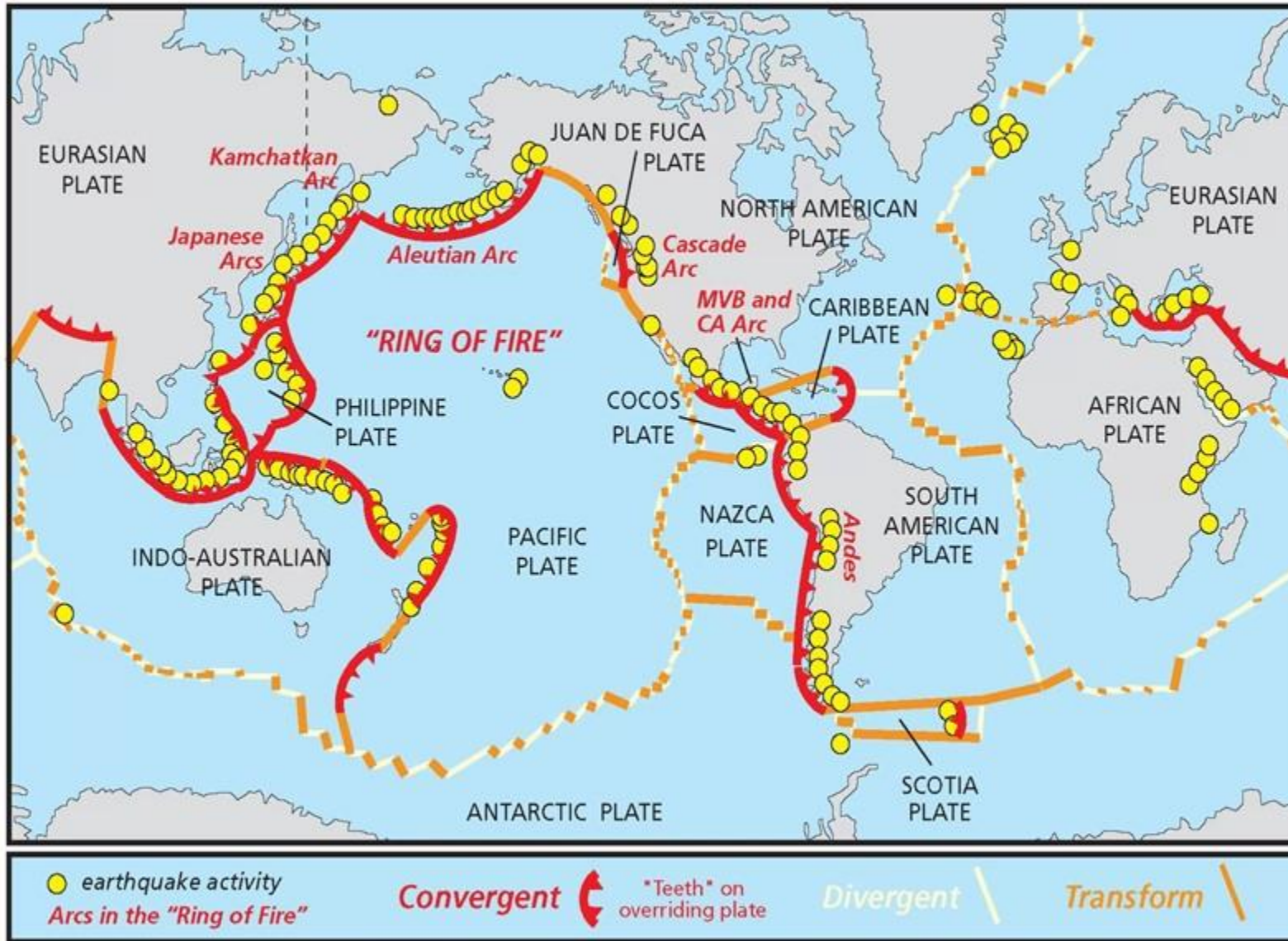
2 This volcanic belt is also called as the **Fire Girdle of the Pacific** or the Fire Ring of the Pacific.

3 This belt begins from Erebus Mountain of Antarctica and runs northward through Andes and Rockies mountains of South and North Americas, eastern Asiatic coast which includes the volcanoes of island arcs and festoons (e.g. Sakhalin, Kamchatka, Japan, Philippines etc.)

RING OF FIRE



EAST-PACIFIC RISE



DISTRIBUTION OF VOLCANOES IN THE WORLD

2 MID-CONTINENTAL BELT

1 Mid-continental belt is also known as 'the volcanic zones of convergent continental plate margins'.

2 This belt includes the volcanoes of the Mediterranean sea and the volcanoes of fault zone of eastern Africa.

3 The famous volcanoes of the Mediterranean sea such as Stromboli, Visuvius, Etna etc belong to this belt.

4 The volcanoes of Aegean sea are included in this belt. It may be pointed out that this belt does not have the continuity of volcanic eruptions as several gaps (volcanic-free zones) are found along the Alps and the Himalayas as because of compact and thick crust formed due to intense folding activity.

MID-CONTINENTAL BELT

1

[illegible]

DISTRIBUTION OF VOLCANOES IN THE WORLD

3

MID-ATLANTIC BELT



1 Mid-Atlantic belt includes the volcanoes mainly along the mid-Atlantic ridge which represents the splitting zone of plates.

2 In other words, two plates diverge in opposite directions from the mid-oceanic ridge.

3 Thus, volcanoes mainly of **fissure eruption type** occur along the constructive or divergent plate margins (boundaries).

4 The most **active volcanic area** is **Iceland**, which is located on the mid-Atlantic ridge





CANADA

UNITED STATES

RUSSIA

North Atlantic Ocean

Madeira

Canary Islands

Cape Verde Islands

South Atlantic Ocean

Georgetown

St. Helena

Indian Ocean

AUSTRALIA

DISTRIBUTION OF VOLCANOES IN THE WORLD

4 INTRAPLATE VOLCANOES

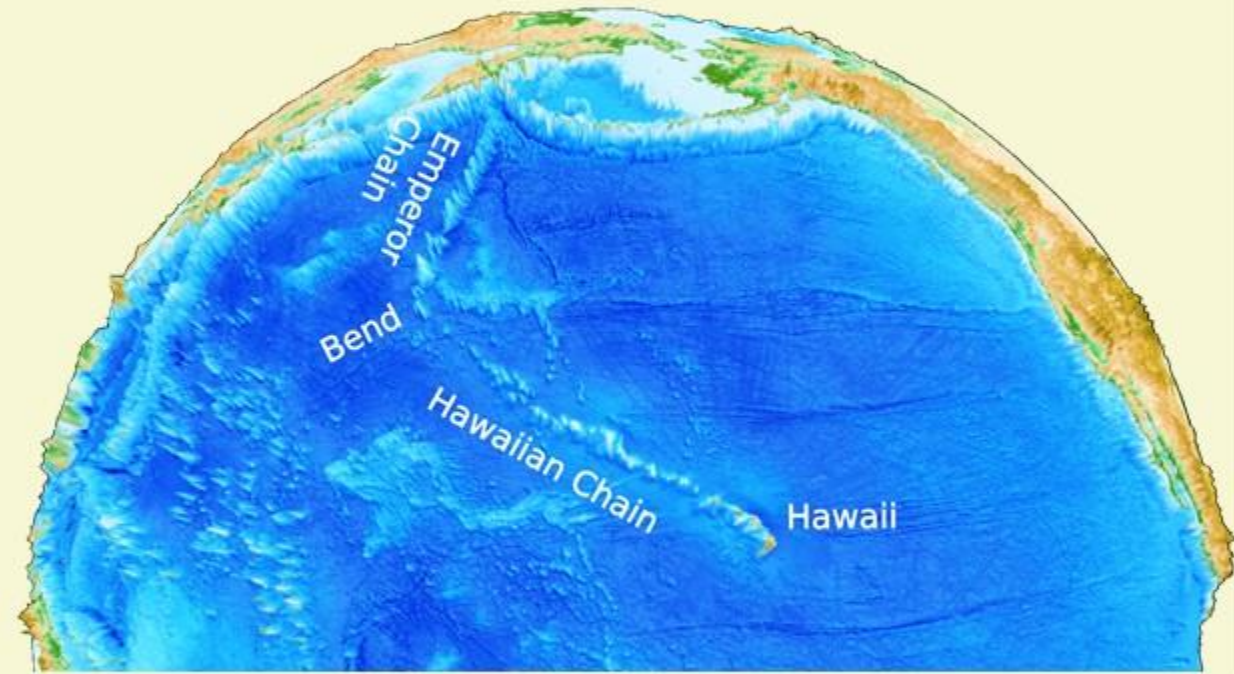
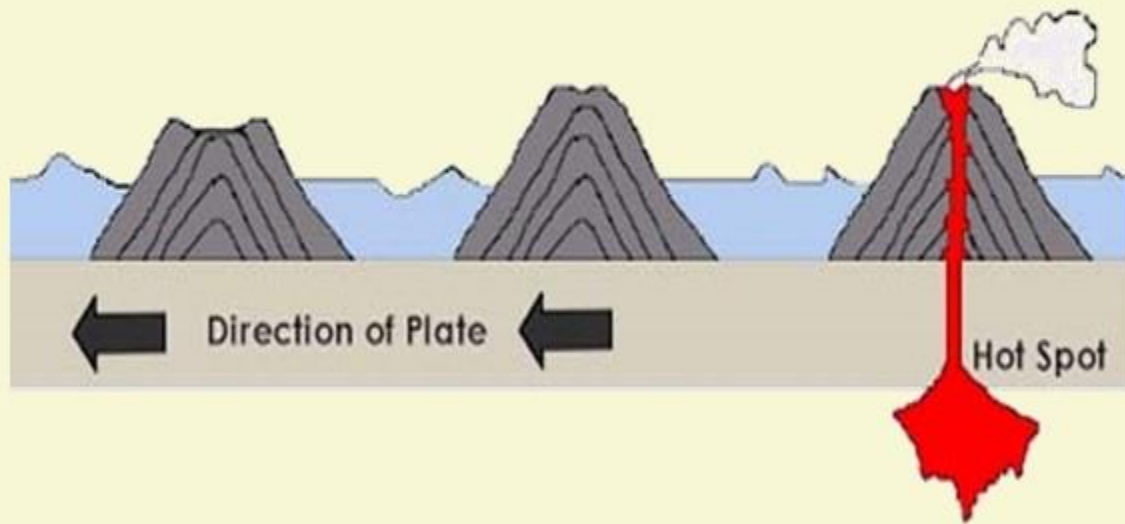


1 Besides, the afore-said well defined three zones of volcanoes, scattered volcanoes are also found in the inner parts of the continents.

2 Such distributional patterns of volcanoes are called as intraplate volcanoes, the mechanism of their eruption is not yet precisely known.

DISTRIBUTION OF VOLCANOES IN THE WORLD

1 INTRAPLATE VOLCANOES



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